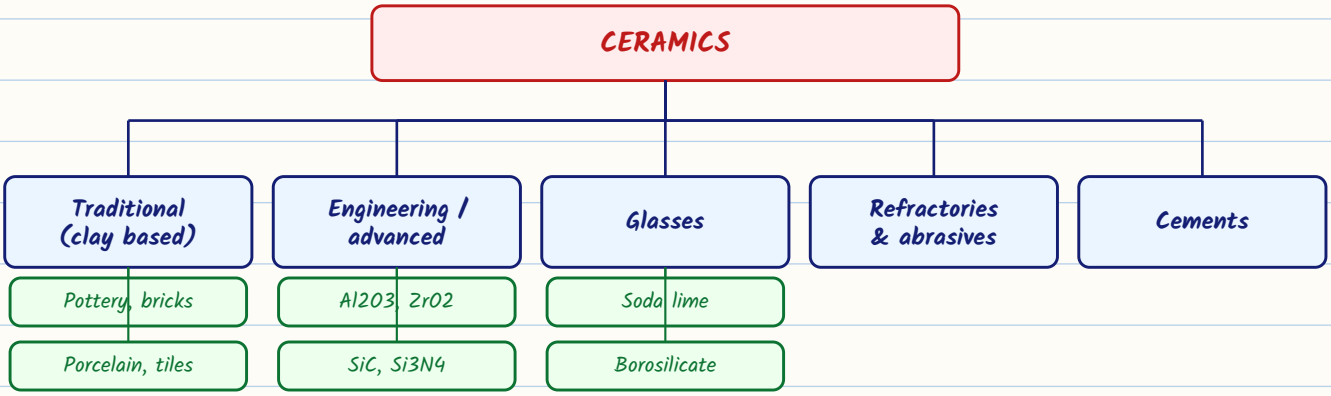
UNIT - V : CERAMIC AND ADVANCED MATERIALS

1. Ceramics - introduction

* CERAMICS = inorganic, non-metallic solids (compounds of metals with O, C, N, B, Si - oxides, carbides, nitrides, borides) processed at high temperature. Bonding is IONIC and / or COVALENT.
* General properties: high hardness and compressive strength, very high melting point, chemical inertness / corrosion resistance, wear resistance, electrical and thermal INSULATION (some are semi / super conductors), low density, but BRITTLE (low tensile strength and toughness) and poor thermal shock resistance.
* Why brittle? Ionic / covalent bonds are directional, very few slip systems; like-charged ions cannot slide past each other; pores and micro-cracks act as stress raisers (Griffith flaw: $\sigma_f = (2 E \gamma / \pi a)^{1/2}$).



2. Crystalline Ceramics

* Crystal structure follows ionic radius ratio r_c/r_a (cation / anion) and charge neutrality. Cations are generally smaller than anions and fit in the interstices of closely packed anions.

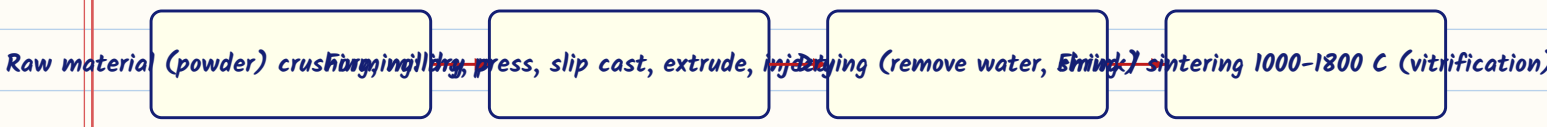
r(c)/r(a)	Coordination number	Structure examples
< 0.155	2 (linear)	-
0.155 - 0.225	3 (triangular)	B2O3
0.225 - 0.414	4 (tetrahedral)	ZnS (zinc blende), SiO2
0.414 - 0.732	6 (octahedral)	NaCl (rock salt), MgO, FeO
0.732 - 1.0	8 (cubic)	CsCl, CaF2 (fluorite) ZrO2

* AX type: rock salt (NaCl, MgO, NiO), CsCl, zinc blende. AmXp: fluorite CaF2 (UO2, ZrO2). AmBnXp: perovskite BaTiO3 (ABO3; piezo / ferro-electric, capacitors), spinel MgAl2O4 (AB2O4; ferrites - magnetic). Silicates: SiO4 tetrahedra (silica, quartz, clay minerals, mica, feldspar).

Important engineering ceramics

Ceramic	Properties	Uses
Alumina Al2O3	Hard (HV 1800), strong, insulator, m.p. 2050 C, chemically stable	Spark plug insulator, substrates, cutting tools, bio-implants, refractory
Zirconia ZrO2 (PSZ, YSZ)	High toughness (transformation toughening), low k, ionic conductor	Thermal barrier coating, oxygen sensors, knives, dental crowns, fuel cells
Silicon carbide SiC	Very hard, strength at 1500 C, high k, oxidation resistant	Abrasives, kiln furniture, heating elements, brake discs, armour
Silicon nitride Si3N4	High strength, thermal shock resistance, creep resistance	Bearings, turbine rotors, cutting tools, engine parts
Boron carbide B4C, BN, TiC, TiB2	Extremely hard, light	Armour, nozzles, neutron absorbers, cutting tools
Beryllia BeO	High thermal conductivity, electrical insulator	Electronic substrates (toxic dust)
Barium titanate BaTiO3, PZT	Ferro / piezo electric, high permittivity	Capacitors, transducers, sensors

Processing of ceramics



* Fired ceramic: crystalline grains + glassy phase + pores (porosity reduces strength). Hot pressing and HIP get dense, strong ceramics. Sol-gel and CVD used for films / fibres.

3. Glasses

* GLASS = inorganic AMORPHOUS (non-crystalline) solid obtained by cooling a melt so fast that crystallisation is avoided; a rigid super-cooled liquid. No sharp melting point - softens gradually over a range (Glass transition temperature T_g).
* Structure: random network of SiO4 tetrahedra (network former: SiO2, B2O3, P2O5), network modifiers (Na2O, K2O, CaO, BaO - break the Si-O-Si bonds, lower viscosity and m.p.), intermediates (Al2O3, PbO, TiO2).
* Properties: transparent, hard, brittle, good chemical resistance (except HF), electrical insulator, low thermal conductivity, isotropic, easily shaped when hot (viscous). Characteristic temperatures: strain point, annealing point (~ 10^{13} poise), softening, working and melting points.

Type of glass	Composition	Properties and uses
Soda-lime (window, bottle)	70-72 SiO2, 12-16 Na2O, 10-12 CaO	Cheap (90% of all glass); windows, bottles, bulbs; poor thermal shock
Borosilicate (Pyrex)	80 SiO2, 13 B2O3, 4 Na2O, 2 Al2O3	Low expansion, thermal shock and chemical resistant; lab ware, cookware, tube lights
Lead glass (crystal)	50-60 SiO2, 25-30 PbO, K2O	High refractive index, brilliance; optics, tableware, radiation shielding
Fused silica / vitreous silica	99.9% SiO2	Very low expansion, UV transparent; crucibles, optical fibre, telescope mirrors
96% silica (Vycor)	96 SiO2, 4 B2O3	High temperature use up to 900 C
Glass ceramic (Pyroceram)	Li2O-Al2O3-SiO2 + TiO2 nucleant	Controlled crystallisation (90% crystalline), strong, zero expansion; cooktops, missile radomes
Toughened / tempered	Soda-lime heat treated	Surface compressive stress; car side windows, doors
Laminated (safety)	Glass + PVB interlayer	Wind screens - does not shatter

* Manufacturing: melting (1400 - 1600 C) -> forming by blowing, pressing, drawing, float process (glass floated on molten tin bath - flat sheets), fibre drawing -> ANNEALING (remove residual stresses) -> finishing.
* Tempering: heat above T_g (about 650 C) then cool surface quickly with air jets - surface in compression, core in tension - 4-5 times stronger. Chemical strengthening: K+ ion exchange.

4. Cermets

* CERMET = CERamic + METal composite: hard refractory ceramic particles bonded with metallic binder; combine the hardness, high m.p. and oxidation resistance of ceramics with the toughness and thermal shock resistance of

metals. Made by powder metallurgy (liquid phase sintering).

Cermet	Constituents	Uses
Cemented carbide	WC + 6-25% Co (also TiC, TaC + Ni)	Cutting tool inserts, drill bits, dies, wear parts
Oxide cermet	Al2O3 + Cr (or Fe, Ni, Mo)	Turbine blades, high temperature parts, nozzles
Titanium carbide cermet	TiC + Ni / Mo	High speed finishing tools, cutting
Nuclear cermets	UO2 + Stainless steel / Mo	Fuel elements
Cr3C2 + Ni (Cr)	Chromium carbide + nichrome	Corrosion resistant, coating by thermal spray
MgO / Al2O3 + Ni	Oxides + metals	Brake linings, electrical contacts

* Cermet properties: hardness 1200-1800 HV, compressive strength very high, density 6-15 g/cc, retain hardness at 800-1000 C. Brittle compared with metals but tougher than monolithic ceramics.

5. Abrasive Materials

* ABRASIVES are hard, tough, sharp-edged materials used for cutting, grinding, polishing and lapping other materials (grinding wheels, coated abrasives - sand paper, abrasive powders, lapping paste, blasting).
* Requirements: high hardness (greater than workpiece), toughness, friability (self-sharpening: fractures to expose new sharp edges), thermal stability, chemical inertness.

Abrasive	Hardness (Knoop / Mohs)	Notes and use
Diamond (natural / synthetic)	7000-8000 / 10	Hardest; grinding carbide, glass, ceramics, dressing wheels, polishing; reacts with Fe at high T
Cubic boron nitride (CBN, Borazon)	4500 / 9.9	Grinds hardened steel, HSS, super alloys; stable up to 1300 C
Silicon carbide (Carborundum)	2500 / 9.5	Grinds cast iron, non-ferrous, glass, stone, ceramic - very sharp but brittle (green or black)
Aluminium oxide (alumina, Corundum / emery)	2000 / 9	Most used; steel and alloy grinding; tougher; white, brown, pink (Cr), sol-gel alumina
Boron carbide	2800 / 9.3	Lapping, nozzles, dressing tools
Garnet, emery, quartz, flint, pumice	800-1200	Natural; sand paper, polishing, sandblasting

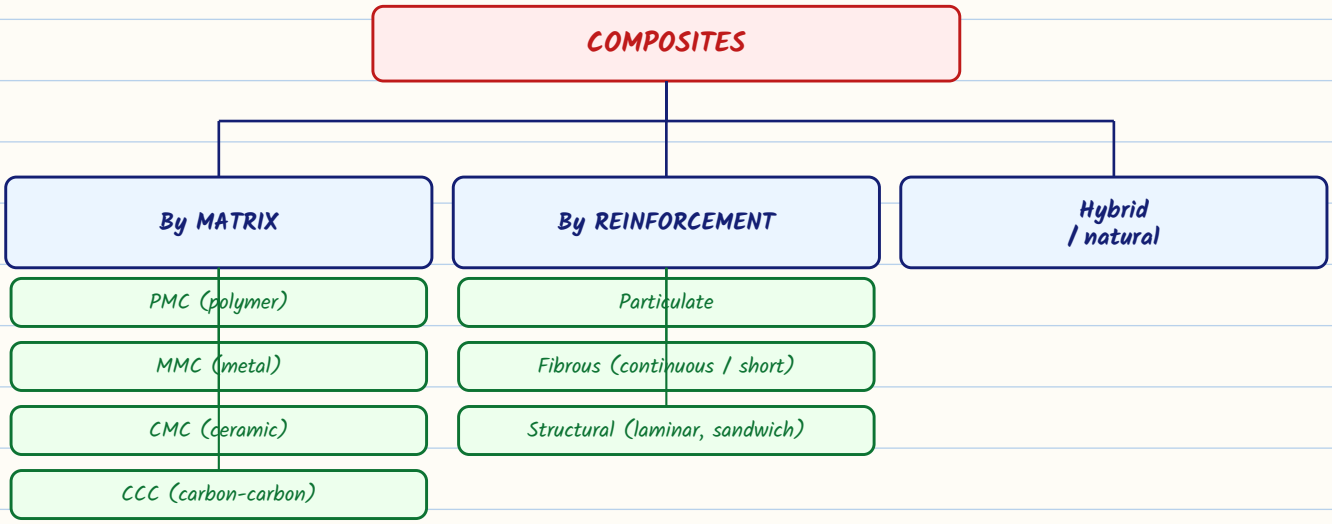
* GRINDING WHEEL: abrasive grains + BOND (vitrified (ceramic clay) - most common, silicate, shellac, rubber, resinoid (bakelite), metal bond (Cu, Ni) for diamond). Wheel marking e.g. A 60 K 5 V: abrasive type, grit size, grade (hardness), structure, bond.

6. Composite Materials

* COMPOSITE = multiphase material of two or more physically distinct and chemically different constituents - the MATRIX (continuous, binds, transfers load, protects) and the REINFORCEMENT (dispersed phase, carries most load) - separated by an interface, giving a combination of properties better than either alone.
* Examples: wood, bone, concrete, fibre glass, plywood, tyre, carbon-epoxy aircraft parts, WC-Co tool.

* Merits: high specific strength and stiffness (strength/density), design flexibility (anisotropy tailored), fatigue and corrosion resistance, light weight, damping, thermal stability. Demerits: costly, anisotropic, difficult to repair, poor out-of-plane strength, delamination, recycling issues.

Classification of composites





Particle reinforced | Fibre reinforced | Structural (laminar)

7. Particle Reinforced Composites

* Reinforcement is hard particles (equiaxed, size 0.01 - 100 micron) in matrix. LARGE particle composites: particle - matrix interaction on macroscale; load shared: concrete (gravel in cement), cermets (WC-Co), carbon black in rubber (tyre), Al-SiC, Ni-Al2O3.

* DISPERSION STRENGTHENED composites: very small particles (10 - 100 nm), 1 - 15 vol% hard oxide (e.g. ThO2 in Ni, Al2O3 in Al (SAP), Y2O3 in super alloys). They block dislocations (Orowan mechanism); matrix carries load; retains strength at high temperature.
* Rule of mixtures for large particles - elastic modulus bounds:

$$\text{Upper: } E_c = E_m V_m + E_p V_p \quad \text{Lower: } 1/E_c = V_m/E_m + V_p/E_p$$

* Uses: tool bits, tyres, grinding wheels, electrical contacts, rocket nozzles, concrete.

8. Fibre Reinforced Composites

* Most important class: strong stiff fibres in a ductile matrix. Fibre is the main load carrier. Mechanical properties depend on (a) fibre-matrix BOND, (b) fibre length (critical length $l_c = \sigma_f d / 2\tau_c$), (c) fibre orientation and (d) fibre volume fraction.
* CONTINUOUS (aligned) fibres: maximum strength along fibre axis (longitudinal), very weak transverse. DISCONTINUOUS (chopped) aligned or random (2-D) - isotropic in plane (sheet moulding compound, SMC). Woven fabrics, mats, 3-D weaves, laminates [0 / 90 / +-45] to give quasi isotropic properties.

Fibres

Fibre	Density g/cc	Tensile strength GPa	Modulus GPa	Remarks
E-glass	2.55	3.4	72	Cheapest, most used (GFRP)
S-glass	2.5	4.5	85	Stronger, aerospace
Carbon (high strength)	1.8	3.5-5.5	230-300	High stiffness, light, brittle (CFRP)
Aramid (Kevlar 49)	1.45	3.6	130	Tough, impact, bullet-proof vests
Boron	2.6	3.5	400	Very stiff, costly (W core)
SiC / alumina	2.5-3.2	2-3.4	200-400	High temperature MMC, CMC
Natural (jute, sisal, flax)	1.3-1.5	0.4-0.8	20-50	Low cost, biodegradable

Rule of mixtures (continuous aligned fibres)

$$E_{cl} = E_f V_f + E_m V_m \text{ (iso-strain, longitudinal)}$$

$$1/E_{c-} = V_f/E_f + V_m/E_m \text{ (iso-stress, transverse)}$$

Eg: 60% E-glass (72 GPa) in epoxy (3 GPa): E_longitudinal = 0.6 x 72 + 0.4 x 3 = 44.4 GPa; transverse: 1/E = 0.6/72 + 0.4/3 = 0.1417 -> E = 7.06 GPa.

9. Polymer, Metal, Ceramic and Carbon-Carbon Composites

(a) PMC - Polymer Matrix Composites

* Matrix: thermosets (epoxy, polyester, phenolic, vinyl ester, polyimide) or thermoplastics (PEEK, nylon, PP, PPS). Reinforcement: glass (GFRP), carbon (CFRP), aramid (AFRP). Service temperature limited (up to 150-300 C).
* Manufacture: HAND LAY-UP / wet lay-up + spray-up (boats, tanks);

VACUUM BAGGING and autoclave moulding of prepregs (aircraft); FILAMENT WINDING (pipes, pressure vessels, rocket motor cases); PULTRUSION (continuous constant section rods, beams, ladders); RESIN TRANSFER MOULDING (RTM, VARTM); compression moulding of SMC / BMC (auto parts); injection moulding (short fibre thermoplastics); prepreg tape laying.

* Uses: aircraft wings and fuselage (Boeing 787 > 50% composite), wind turbine blades, car bodies, boats, sports goods (rackets, golf shafts, bicycles), pipes, printed circuit boards, tanks.

(b) MMC - Metal Matrix Composites

* Matrix: Al, Mg, Ti, Cu, Ni (ductile) reinforced by SiC, Al_2O_3, B_4C particles or whiskers, carbon, boron or SiC fibres. Higher service temperature (to 500 C), higher strength, stiffness, wear resistance, thermal / electrical conductivity than PMC, no moisture absorption; but costlier and heavier.
* Manufacture: STIR CASTING (particles stirred into molten Al, then cast - cheapest), SQUEEZE CASTING / liquid infiltration, POWDER METALLURGY (blend, press, sinter), diffusion bonding (fibre + foil), spray deposition, in-situ reactions, friction stir processing.
* Uses: Al-SiC brake discs and drums, piston and cylinder liners, connecting rods, drive shafts, bicycle frames, electronic heat sinks, aerospace structures (Al-B), turbine blades.

(c) CMC - Ceramic Matrix Composites

* Matrix: Al_2O_3, SiC, Si_3N_4, glass-ceramic, mullite, carbon; reinforcement: SiC fibres / whiskers, carbon fibre, Al_2O_3 fibres, zirconia particles. Fibres are added NOT for strength but to increase TOUGHNESS - crack deflection, fibre pull-out and bridging prevent catastrophic failure.
* Manufacture: hot pressing, CVI (chemical vapour infiltration), reaction bonding, sol-gel, slurry infiltration + sintering, polymer infiltration pyrolysis (PIP), melt infiltration. Uses: jet engine hot sections (shrouds, combustor liners), brake discs (C/SiC in Porsche, supercar), heat shields, cutting tools (Al_2O_3-SiC whisker), armour, gas turbine blades.

(d) CCC - Carbon-Carbon Composites

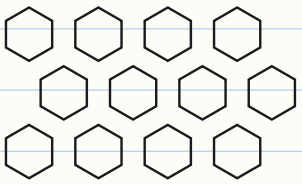
* Carbon fibres in a carbon (graphite) matrix. Made by laying fibres (2-D, 3-D weave) with resin / pitch -> carbonise (800-1000 C) -> repeated DENSIFICATION by impregnation + pyrolysis or CVI -> graphitise (2500 C+).
* Properties: strength and stiffness maintained up to 2000+ C (in inert / protected atmosphere), very low density (1.6-1.9 g/cc), low thermal expansion, high thermal shock resistance, good friction and wear, biocompatible; weakness: OXIDATION above 400-500 C (needs SiC coating).
* Uses: space shuttle nose cone and wing leading edges, rocket nozzles and throats, aircraft and F1 racing car brake discs, hot press dies, furnace fixtures, bone plates and implants.

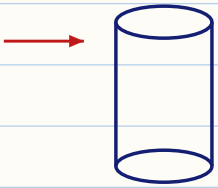
Property	PMC	MMC	CMC	CCC
Max temp	150-300 C	300-600 C	1000-1600 C	2000+ C (inert)
Density	Lowest	Medium	Medium	Low
Toughness	Good	Good	Poor -> improved	Fair
Cost	Low-medium	Medium-high	High	Highest

10. Introduction to Nanomaterials

* NANOMATERIALS: materials with at least ONE dimension in the nanometre range (1 - 100 nm; 1 nm = 10^{-9} m) - properties differ from bulk due to (i) very large SURFACE AREA / volume ratio, (ii) quantum confinement effects.
* Classification by dimension: 0-D (nanoparticles, quantum dots, fullerene C60), 1-D (nanowires, nanotubes, nanorods), 2-D (thin films, graphene, nanoplates), 3-D (nanocrystalline bulk, nanocomposites).

Graphene sheet Rolled -> nanotube





SWCNT dia 0.4-3 nm,
MWCNT 2-100 nm
Tensile strength > 50 GPa
E about 1 TPa, k = 3000 W/mK
Armchair - metallic;
zigzag / chiral - semiconducting

Types and properties

* FULLERENE C60 (buckyball; Kroto, Smalley, Curl 1985): 60 carbon atoms in a soccer ball (20 hexagons, 12 pentagons). CARBON NANOTUBES (Iijima 1991): rolled graphene sheet: SWCNT / MWCNT, very high strength (50-100x steel at 1/6 weight), high conductivity. GRAPHENE: single layer of C atoms (2004, Geim, Novoselov), 2-D, strongest known, very high carrier mobility. Quantum dots (CdSe) - size dependent colour. Nano-Ag, TiO2, ZnO, Au particles.
* Size effects: Hall-Petch strengthening (nanocrystalline metals are 3-5 times harder), melting point falls, ceramics become ductile (superplastic), optical colour changes (Au nanoparticles red), magnetic: superparamagnetism, catalytic activity rises with surface.

Synthesis

* TOP-DOWN: ball milling (mechanical alloying), lithography, etching, severe plastic deformation (ECAP). BOTTOM-UP: chemical vapour deposition (CVD), sol-gel, co-precipitation, laser ablation, arc discharge (CNT, fullerene), electro-deposition, self assembly, inert gas condensation, hydrothermal.

Applications

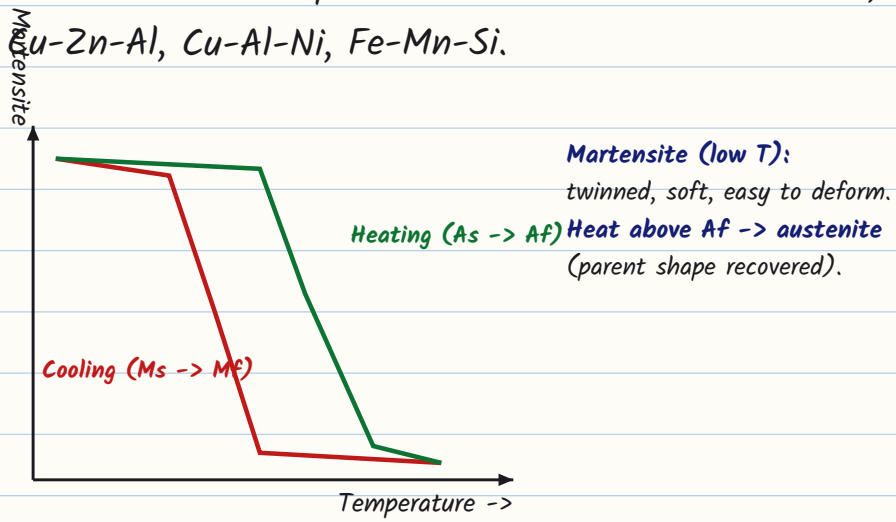
* Electronics (transistors, memory, displays, sensors), medicine (drug delivery, imaging, cancer therapy, antibacterial Ag), energy (solar cells, Li-ion batteries, fuel cells, H2 storage), cosmetics (sunscreen ZnO / TiO2), water purification, catalysts, nano-composites (sports goods), coatings (self cleaning, scratch resistant), textiles. Concerns: toxicity, environmental impact.

II. Smart Materials

* SMART (intelligent / responsive) materials sense a change in environment (stimulus: temperature, stress, electric / magnetic field, light, pH) and respond by a CHANGE of one or more properties (shape, colour, stiffness, viscosity) in a reversible and controllable way. Used as sensors, actuators and adaptive structures.

(a) Shape memory alloys (SMA)

* Alloys that 'remember' original shape: after plastic deformation at low temperature (martensite) they return to the shape on heating above transformation temperature (austenite). NiTi (Nitinol, 55 Ni - 45 Ti), Cu-Zn-Al, Cu-Al-Ni, Fe-Mn-Si.



* Thermoelastic martensite transformation: one-way effect (shape recovered only on heating), two-way effect (trained for both shapes), SUPER-ELASTICITY (pseudo-elasticity: 8% recoverable strain above Af, by stress induced martensite). Uses: orthodontic wires, stents, eye-glass frames, couplings and fasteners, robotic actuators, thermostat valves, aircraft wing morphing, anti-scald valves.

(b) Piezoelectric materials

* Generate voltage when mechanically stressed (direct effect) and deform on applying electric field (converse effect). Quartz, Rochelle salt, BaTiO3, PZT (lead zirconate titanate), PVDF polymer. Uses: ultrasonic transducers, sensors, microphone, lighter, sonar, inkjet, actuators, energy harvesting, accelerometers,

quartz oscillators.

(c) Magnetostrictive and magnetic SMA

* Terfenol-D (Tb-Dy-Fe) changes length in a magnetic field (up to 0.2%); used in sonar, vibration control, precision actuators. Ferromagnetic shape memory alloys (Ni-Mn-Ga) respond to field.

(d) Electro- and magneto- rheological fluids (ER / MR)

* Suspensions of fine particles in oil that change from liquid to near-solid (viscosity up to 10^5 times in ms) when electric or magnetic field is applied. Uses: adaptive shock absorbers and dampers, clutches, brakes, car suspension, prosthetic limbs, seismic dampers.

(e) Others

* PHOTOCHROMIC (glass with AgCl - darkens in UV, spectacles) and ELECTROCHROMIC (smart windows, mirrors) materials; THERMOCHROMIC (colour changes with T - thermometers); Fibre optic sensors; Self healing polymers and concrete; Electro-active polymers (artificial muscles); Hydrogels (pH / temperature responsive); Thermoelectric materials (Bi2Te3: heat <-> electricity); Chemo-chromic.

Smart material	Stimulus	Response	Application
SMA (Nitinol)	Temperature / stress	Shape change	Stents, wires, actuators
Piezoelectric (PZT)	Stress / Electric field	Voltage / strain	Sensors, sonar
Magnetostrictive (Terfenol-D)	Magnetic field	Length change	Actuators
MR / ER fluid	Magnetic / electric field	Viscosity	Dampers, clutches
Photochromic glass	UV light	Colour	Sunglasses

Revision: Ceramics - ionic / covalent, hard, brittle. Glass - amorphous. Cermet - WC + Co. Composites = matrix + reinforcement; PMC, MMC, CMC, CCC. Nano: 1-100 nm, CNT, graphene. Smart: SMA (Nitinol), piezo (PZT), MR fluid.